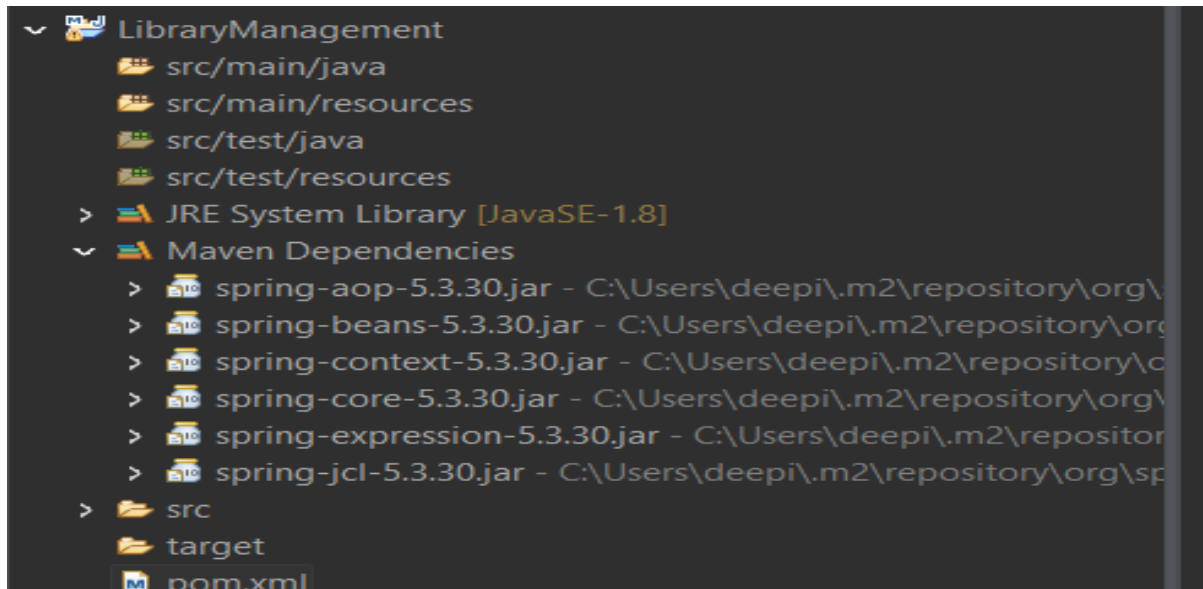
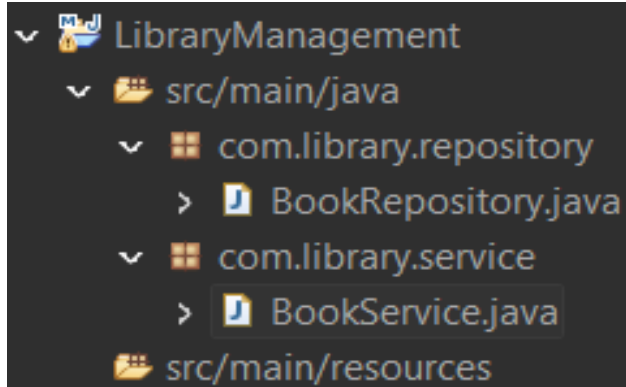


SPRING CORE AND MAVEN

Exercise 1: Configuring a Basic Spring Application



```
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2  <modelVersion>4.0.0</modelVersion>
3  <groupId>com.library</groupId>
4  <artifactId>LibraryManagement</artifactId>
5  <version>0.0.1-SNAPSHOT</version>
6
7  <dependencies>
8
9      <!-- Spring Context -->
10     <dependency>
11         <groupId>org.springframework</groupId>
12         <artifactId>spring-context</artifactId>
13         <version>5.3.30</version>
14     </dependency>
15
16 </dependencies>
17 </project>
```



```
1 package com.library.repository;
2
3 public class BookRepository {
4
5     public BookRepository() {
6         System.out.println("BookRepository Object Created");
7     }
8
9     public void displayRepository() {
10         System.out.println("BookRepository is working...");
11     }
12
13 }
```

```
1 package com.library.service;
2
3 public class BookService {
4
5     public BookService() {
6         System.out.println("BookService Object Created");
7     }
8
9     public void displayService() {
10         System.out.println("BookService is working...");
11     }
12
13 }
```

```

1 package com.library.repository;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.library.repository.BookRepository;
7 import com.library.service.BookService;
8
9 public class LibraryManagementApplication {
10
11     public static void main(String[] args) {
12
13         ApplicationContext context =
14             new ClassPathXmlApplicationContext("applicationContext.xml");
15
16         BookRepository repository =
17             context.getBean("bookRepository", BookRepository.class);
18
19         BookService service =
20             context.getBean("bookService", BookService.class);
21
22         repository.displayRepository();

```

```

22         repository.displayRepository();
23
24         service.displayService();
25
26     }
27
28 }
29

```

Console ×

<terminated> LibraryManagementApplication [Java Application] C:\Program Files\Java\jdk-22\bin\javaw

BookRepository Object Created
 BookService Object Created
 BookRepository is working...
 BookService is working...

Exercise 2: Implementing Dependency Injection

```
https://www.springframework.org/schema/beans/spring-beans.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2
3• <beans xmlns="http://www.springframework.org/schema/beans"
4     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5     xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9•     <bean id="bookRepository"
10         class="com.library.repository.BookRepository"/>
11
12•     <bean id="bookService"
13         class="com.library.service.BookService">
14         <property
15             name="bookRepository"
16             ref="bookRepository"/>
17     </bean>
18
19 </beans>
```

```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6
7     private BookRepository bookrepository;
8•     public BookService() {
9         System.out.println("BookService Object Created");
10     }
11
12     //Setter method
13•     public void setBookRepository(BookRepository bookrepository) {
14         this.bookrepository = bookrepository;
15     }
16
17•     public void displayService() {
18         System.out.println("BookService is working...");
19         bookrepository.displayRepository();
20     }
21
22 }
```

```

1 package com.library.repository;
2
3 public class BookRepository {
4
5     public BookRepository() {
6         System.out.println("BookRepository Object Created");
7     }
8
9     public void displayRepository() {
10         System.out.println("BookRepository is working...");
11     }
12
13 }

```

```

1 package com.library.repository;
2
3 import org.springframework.context.ApplicationContext;
4
5 public class LibraryManagementApplication {
6
7     public static void main(String[] args) {
8
9         ApplicationContext context =
10             new ClassPathXmlApplicationContext("applicationContext.xml");
11
12         BookRepository repository =
13             context.getBean("bookRepository", BookRepository.class);
14
15         BookService service =
16             context.getBean("bookService", BookService.class);
17
18         service.displayService();
19
20     }
21 }

```

Console ×

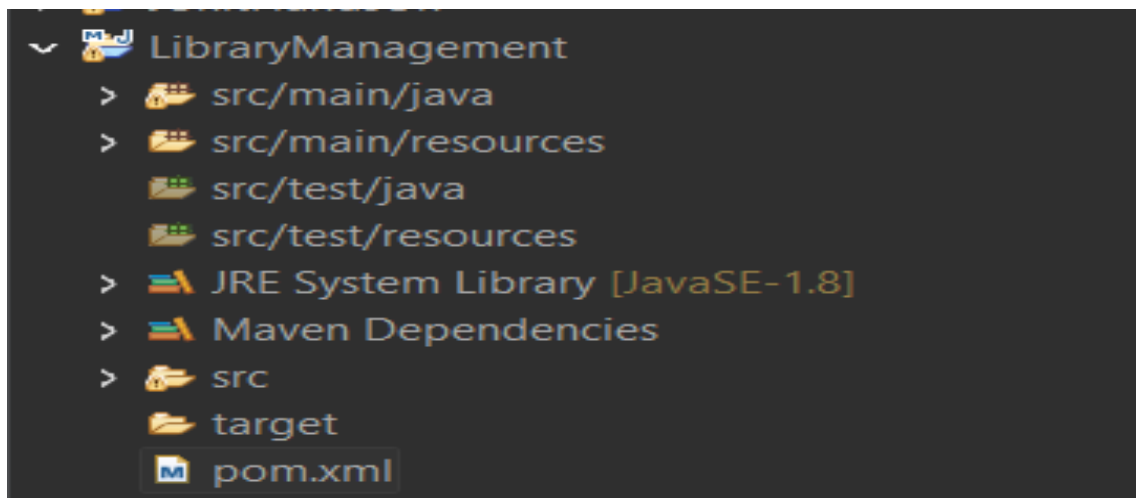
<terminated> LibraryManagementApplication [Java Application] C:\Program Files\Java\jdk

```

BookRepository Object Created
BookService Object Created
BookService is working...
BookRepository is working...

```

Exercise 4: Creating and Configuring a Maven Project



```
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2  <modelVersion>4.0.0</modelVersion>
3  <groupId>com.library</groupId>
4  <artifactId>LibraryManagement</artifactId>
5  <version>0.0.1-SNAPSHOT</version>
6
7  <dependencies>
8
9      <!-- Spring Context -->
10     <dependency>
11         <groupId>org.springframework</groupId>
12         <artifactId>spring-context</artifactId>
13         <version>5.3.30</version>
14     </dependency>
15
16
17     <!-- Spring AOP -->
18     <dependency>
19         <groupId>org.springframework</groupId>
20         <artifactId>spring-aop</artifactId>
21         <version>5.3.30</version>
22     </dependency>
23
24     <!-- Spring WebMVC -->
25     <dependency>
26         <groupId>org.springframework</groupId>
27         <artifactId>spring-webmvc</artifactId>
28         <version>5.3.30</version>
29     </dependency>
```

```

29     </dependency>
30
31 </dependencies>
32
33• <build>
34
35•     <plugins>
36
37•         <plugin>
38
39             <groupId>org.apache.maven.plugins</groupId>
40
41             <artifactId>maven-compiler-plugin</artifactId>
42
43             <version>3.8.1</version>
44
45•         <configuration>
46
47             <source>1.8</source>
48
49             <target>1.8</target>
50
51         </configuration>
52
53     </plugin>
54
55 </plugins>
56
57 </build>
58 </project>

```

Maven Dependencies

- >  spring-aop-5.3.30.jar - C:\Users\deepi\.m2\repository\org\
- >  spring-beans-5.3.30.jar - C:\Users\deepi\.m2\repository\org\
- >  spring-context-5.3.30.jar - C:\Users\deepi\.m2\repository\c\
- >  spring-core-5.3.30.jar - C:\Users\deepi\.m2\repository\org\
- >  spring-expression-5.3.30.jar - C:\Users\deepi\.m2\repositor\
- >  spring-jcl-5.3.30.jar - C:\Users\deepi\.m2\repository\org\sp\
- >  spring-web-5.3.30.jar - C:\Users\deepi\.m2\repository\org\
- >  spring-webmvc-5.3.30.jar - C:\Users\deepi\.m2\repository\